Homework — Nonlinear Equations in One Variable and Systems of Equations in Two Variables

Student: Lavanya Thondavada **Assigned:** 2026-09-10 **Due:** 2026-09-17 **Skills:** Nonlinear equations in one variable and systems of equations in two variables (current), spiral review: Equivalent expressions

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1. [Nonlinear equations · Easy] 3c95093c

$6x - 9y > 12$. Which of the following inequalities is equivalent to the inequality above?

A. $x - y > 2$ B. $2x - 3y > 4$ C. $3x - 2y > 4$ D. $3y - 2x > 2$

2. [Nonlinear equations · Easy] 1e003284

$x = 49$ $y = \sqrt{x} + 9$

The graphs of the given equations intersect at the point (x, y) in the xy-plane. What is the value of y?

A. 16 B. 40 C. 81 D. 130

3. [Nonlinear equations · Medium] 4e18fc5d

$v = -w / (150x)$. The given equation relates the distinct positive numbers v, w, and x. Which equation correctly expresses w in terms of v and x?

A. $w = -150vx$ B. $w = -150v/x$ C. $w = -x/(150v)$ D. $w = v + 150x$

4. [Nonlinear equations · Medium] d0a7871e

$y = x + 1$ $y = x^2 + x$

If (x, y) is a solution to the system of equations above, which of the following could be the value of x?

A. -1 B. 0 C. 2 D. 3

5. [Nonlinear equations · Medium] b4acba95

$x^2 - 12x + 27 = 0$. How many distinct real solutions does the given equation have?

A. Exactly two B. Exactly one C. Zero D. Infinitely many

6. [Nonlinear equations · Hard] 3a9d60b2

$2|4 - x| + 3|4 - x| = 25$. What is the positive solution to the given equation?

(fill in)

7. [Nonlinear equations · Hard] 4661e2a9

$x - y = 1$ $x + y = x^2 - 3$

Which ordered pair is a solution to the system of equations above?

A. $(1+\sqrt{3}, \sqrt{3})$ B. $(\sqrt{3}, -\sqrt{3})$ C. $(1+\sqrt{5}, \sqrt{5})$ D. $(\sqrt{5}, -1+\sqrt{5})$

8. [Equivalent expressions · Easy] f5c3e3b8

Which expression is equivalent to $(m^4q^4z^{-1})(mq^5z^3)$, where m, q, and z are positive?

A. $m^4q^{20}z^{-3}$ B. $m^5q^9z^2$ C. $m^6q^8z^{-1}$ D. $m^{20}q^{12}z^{-2}$

9. [Equivalent expressions · Easy] 72ebc024

Which expression is equivalent to $16x^3y^2 + 14xy$?

A. $2xy(8xy + 7)$ B. $2xy(8x^2y + 7)$ C. $14xy(2x^2y + 1)$ D. $14xy(8x^2y + 1)$

10. [Equivalent expressions · Medium] dd4ab4c4

$4a^2 + 20ab + 25b^2$. Which of the following is a factor of the polynomial above?

A. $a + b$ B. $2a + 5b$ C. $4a + 5b$ D. $4a + 25b$

11. [Equivalent expressions · Medium] 52931bfa

Which expression is equivalent to $(8x(x-7) - 3(x-7)) / (2x - 14)$, where $x > 7$?

A. $(x-7)/5$ B. $(8x-3)/2$ C. $(8x^2-3x-14)/(2x-14)$ D. $(8x^2-3x-77)/(2x-14)$

12. [Equivalent expressions · Hard] 371cbf6b

$(ax+3)(5x^2-bx+4) = 20x^3 - 9x^2 - 2x + 12$. The equation above is true for all x, where a and b are constants. What is the value of ab?

A. 18 B. 20 C. 24 D. 40

Answer Key — Nonlinear Equations Homework (Lavanya Thondavada, 2026-09-10)

#	Skill	Answer	Source	Rationale
1	Nonlinear equations	B	3c95093c	Divide both sides by 3: $2x-3y>4$.
2	Nonlinear equations	A	1e003284	Substitute $x=49$: $y=\sqrt{49}+9=16$.
3	Nonlinear equations	A	4e18fc5d	Multiply both sides by $-150x$: $w=-150vx$.
4	Nonlinear equations	A	d0a7871e	$x+1=x^2+x \rightarrow x^2=1 \rightarrow x=\pm1$; only -1 is a choice.
5	Nonlinear equations	A	b4acba95	Discriminant = $(-12)^2-4(1)(27)=36>0 \rightarrow$ two distinct real solutions.
6	Nonlinear equations	9	3a9d60b2	$5\|4-x\|=25 \rightarrow \|4-x\|=5 \rightarrow x=-1$ or 9; positive solution is 9.
7	Nonlinear equations	A	4661e2a9	Substitution gives $y^2=3$, $y=\sqrt{3}$, $x=1+\sqrt{3}$.
8	Equivalent expressions	B	f5c3e3b8	Add exponents on matching bases: $m^5q^9z^2$.
9	Equivalent expressions	B	72ebc024	GCF is $2xy$: $2xy(8x^2y+7)$.
10	Equivalent expressions	B	dd4ab4c4	Perfect-square trinomial: $(2a+5b)^2$.
11	Equivalent expressions	B	52931bfa	Cancel common factor $(x-7)$: $(8x-3)/2$.
12	Equivalent expressions	C	371cbf6b	Match x^2 coefficients: $-ab+15=-9 \rightarrow ab=24$.